

	1	2	3	4	5	6	7	8	
A	MARV V2 flight controller – schematic index. One page per subsystem; on each page the IC is drawn with its own decoupling, pull–ups, series resistors, test points and connectors wired to it, and a global label is used only where a net leaves the page. See DESIGN_SPEC.md for the design itself. Inputs: regulated 5 V from an EXTERNAL BEC on the MicoAir AM32 ESC pad row J3 pin 7, and USB VBUS. J3 pin 9 brings raw pack VBAT in as a SENSE wire only – no load, no bulk, no copper zone. Rails: 5V_IN is the BEC rail and the servo/module bus; V5_SYS is 5V_IN OR USB through the Q1/D1 OR and feeds only the two 3.3 V regulators and the status LED; V3V3_SYS from the AP63203 buck, V3V3_ANA from the TPS7A20 LDO.								A
B	POWER 1 / ESC pad row, 5 V input and analog sense File: power_vbat.kicad_sch		BOARD 5 / USB–C, D+/D– series pair, SWD pads File: usb_debug.kicad_sch		BOARD 3 / microSD socket File: sd.kicad_sch				B
C	POWER 2 / 5 V OR – BEC P–FET and USB Schottky File: power_or.kicad_sch		BOARD 2 / ICM–45686 six–axis IMU File: imu.kicad_sch		BOARD 3 / optional QSPI expansion socket (DNP) File: qspi_expansion.kicad_sch				C
D	POWER 3 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch		BOARD 2 / BMP581 barometer File: baro.kicad_sch		BOARD 4 / IO array, status LED, mounting holes File: io_block.kicad_sch				D
E	BOARD 1 / RP2354B File: mcu.kicad_sch		BOARD 2 / ADXL375 high–g accelerometer File: highg.kicad_sch						E
F	1	2	3	4	5	6	7	8	

Sheet: /
File: MARV–V2.kicad_sch

Title: MARV–V2 – schematic index

Size: A3

Date: 2026–09–16

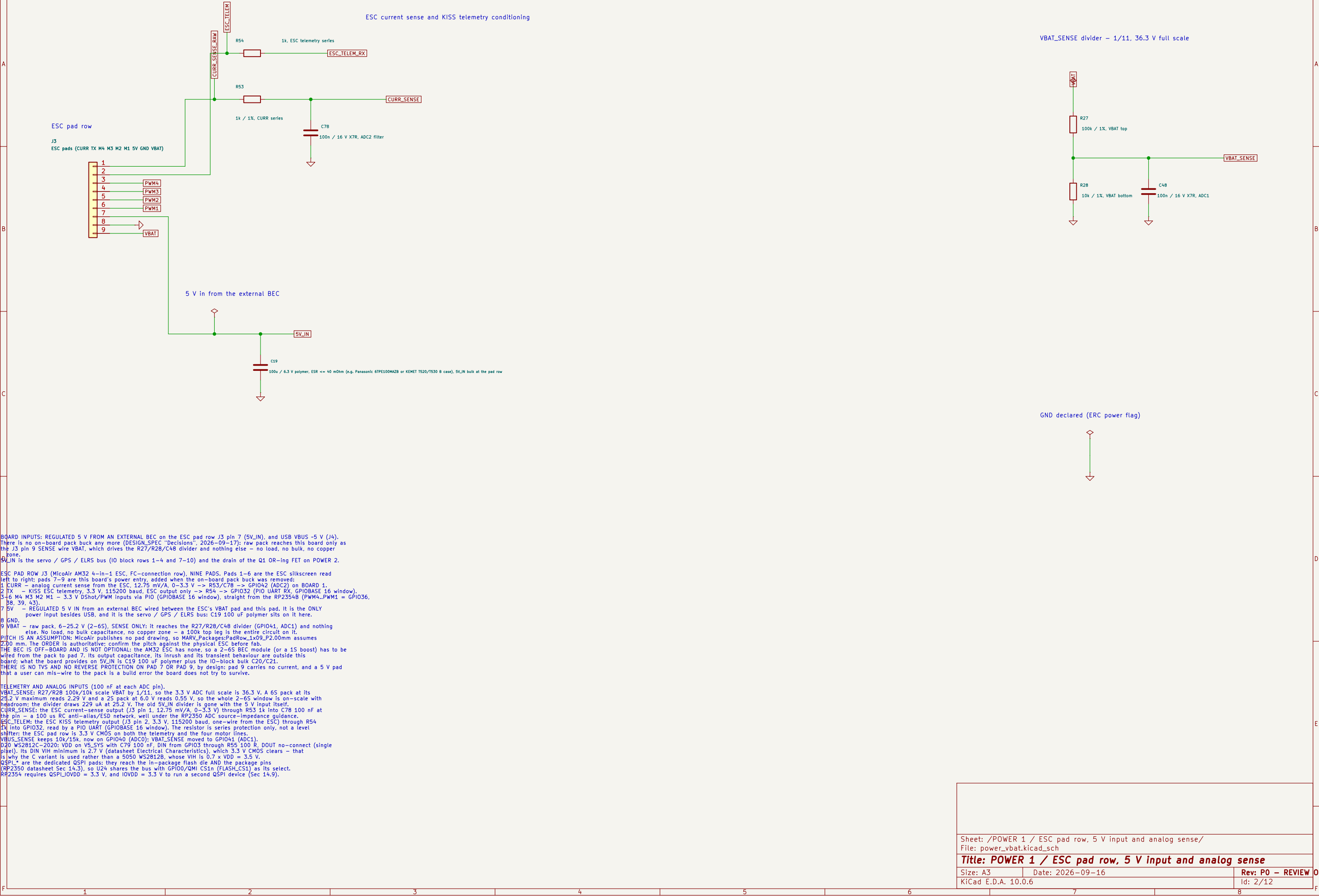
Rev: P0 – NOT FOR RELEASE

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Id: 1/12

Sheet: / File: MARV–V2.kicad_sch		
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POWER 1 – ESC PAD ROW J3 (NINE PADS), 5 V INPUT FROM AN EXTERNAL BEC, VBAT SENSE, ANALOG SENSE.
Regulated 5 V arrives on J3 pin 7 as 5V_IN; pin 9 brings raw pack VBAT in as a SENSE wire only.



POWER 2 – Q1 AO3401A P–FET + D1 1N5819WS SCHOTTKY. 5V_IN (external BEC, J3 pin 7) OR USB_VBUS –> V5_SYS.
USB present turds Q1 OFF, so a host can never back–feed the BEC, the servo rows or the ESC.

A

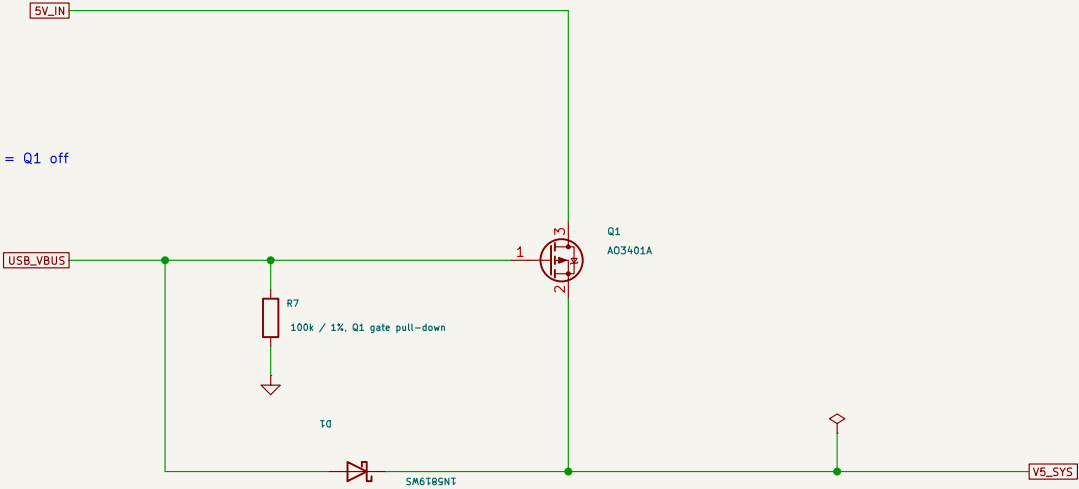
B

C

E

F

BEC path – Q1 P–channel FET, drain 5V_IN, source V5_SYS



USB path – D1 Schottky, the only way USB reaches V5_SYS

V5_SYS – the two 3.3 V regulators and the status LED, nothing else

5 V OR – Q1 P–CHANNEL FET + D1 SCHOTTKY, NOT A MUX (DESIGN_SPEC "Decisions"). 5V_IN (external BEC) and USB_VBUS both reach V5_SYS, and V5_SYS feeds only the two 3.3 V regulators and the status LED.
Q1 AO3401A (AOS, SOT–23, LCSC C15127, JLC BASIC); VDS –30 V, ID –4.0 A, RDS(on) 60 mOhm max at VGS = –4.5 V / 85 mOhm max at –2.5 V, VGS(th) –0.45 to –1.0 V, DRAIN on 5V_IN, SOURCE on V5_SYS, GATE on USB_VBUS with R7 100k to GND.
The intrinsic body diode of a P–channel FET points DRAIN –> SOURCE, i.e. 5V_IN –> V5_SYS, which is the direction the load is in – so the BEC always gets through even before the channel is enhanced.
NO USB: R7 (helped by the R29/R30 VBUS_SENSE divider, 25k to GND) holds the gate at 0 V, so VGS = –V5_SYS ~ –5 V and the channel is hard on; the BEC carries both regulators through <= 60 mOhm.
USB PRESENT: the gate sits at VBUS, so VGS = VBUS – V5_SYS >= 0 and the channel is OFF. That is the one guarantee this OR has to make – the host can never back–feed the BEC, the servo rows or the ESC.
D1 1N5819WS (SOD–323, LCSC C191023, JLC BASIC); ANODE USB_VBUS, CATHODE V5_SYS, 40 V / 1 A Schottky, VF ~0.45 V at the –250 mA V5_SYS draws on USB, so V5_SYS is ~4.4–4.6 V from a 5 V host – above the AP63203 3.8 V minimum input with margin –
and the diode blocks the reverse direction that Q1 no longer has to.
BOTH PRESENT: the BEC wins whenever 5V_IN – IxRDS(on) exceeds VBUS – VF, which at these currents it always does; Q1 is held off in that state, so nothing flows back into the host either way.
WHAT IS GIVEN UP against the TPS2121 this replaces: no priority comparator, no input over–voltage lockout, no current limit, no soft–start, no open–drain status pin (GPIO37 is a plain spare again) and no seamless make–before–break – the crossover is a diode, not a 100 us controlled switchover. TWO JLC BASIC PARTS INSTEAD OF ONE EXTENDED IC PLUS SIX RESISTORS AND THREE CAPACITORS, and no 5 V path exists that USB can drive backwards.
C19 (100 uF polymer) IS ON 5V_IN AT THE PAD ROW, NOT ON V5_SYS: the hold–up belongs where the servo/module rows are, which is ahead of the OR. V5_SYS carries only C8/C9 at the AP63203 input.

USB_VBUS feeds the D1 Schottky and the Q1 gate on POWER 2 directly: no inrush or current limiting on this sheet, and no data ESD array any more – U8 USBLC6–2SC6 was JLC Extended and is removed (DESIGN_SPEC "Decisions"), so D+/D– go straight from J4 into the 27 R series pair R22/R23, as on the madflight FC3v2. The RP2354B USB pins carry their own on–die ESD structures: a board–level array is what this design gives up, deliberately.

Sheet: /POWER 2 / 5 V OR – BEC P–FET and USB Schottky/
File: power_or.kicad_sch

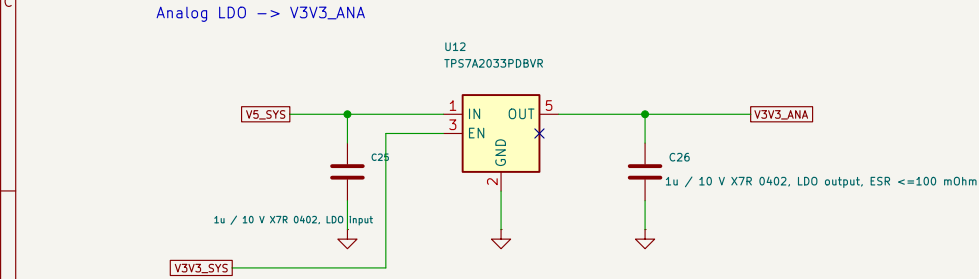
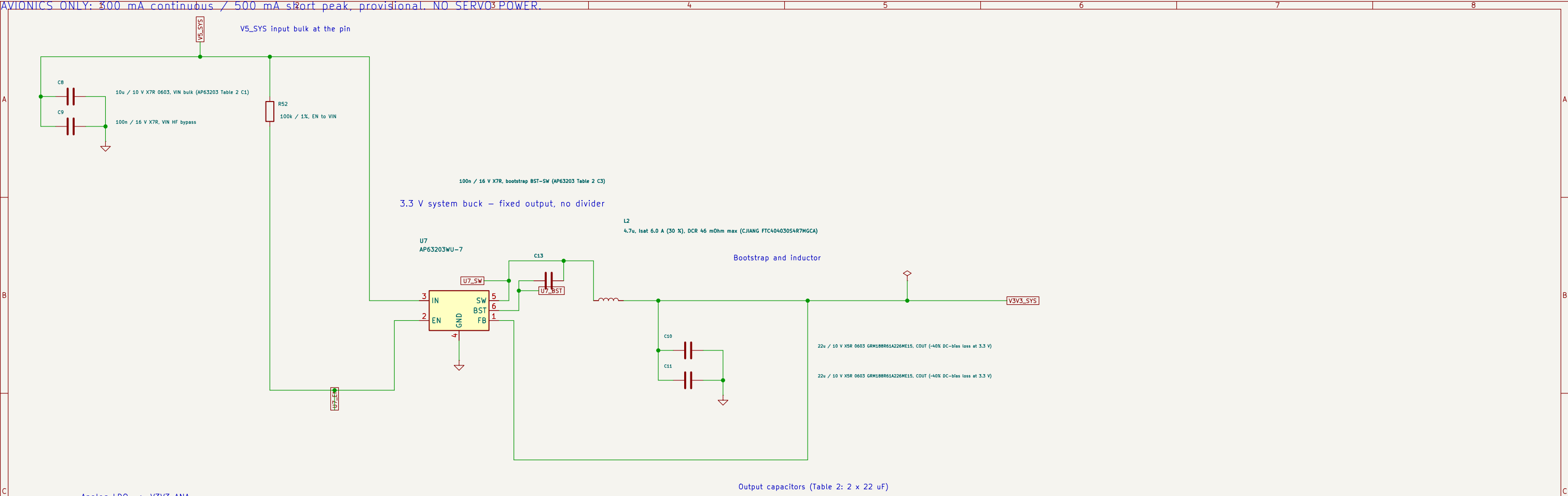
Title: POWER 2 / 5 V OR – BEC P–FET and USB Schottky

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Date: 2026–09–16

Rev: P0 – REVIEW ONLY
Id: 3/12

POWER 3 – U7 AP63203WU–7 FIXED 3.3 V BUCK (V5_SYS –> V3V3_SYS) AND U12 TPS7A20 LDO (–> V3V3_ANA)
AVIONICS ONLY: 300 mA continuous / 500 mA start peak, provisional, NO SERVO POWER.

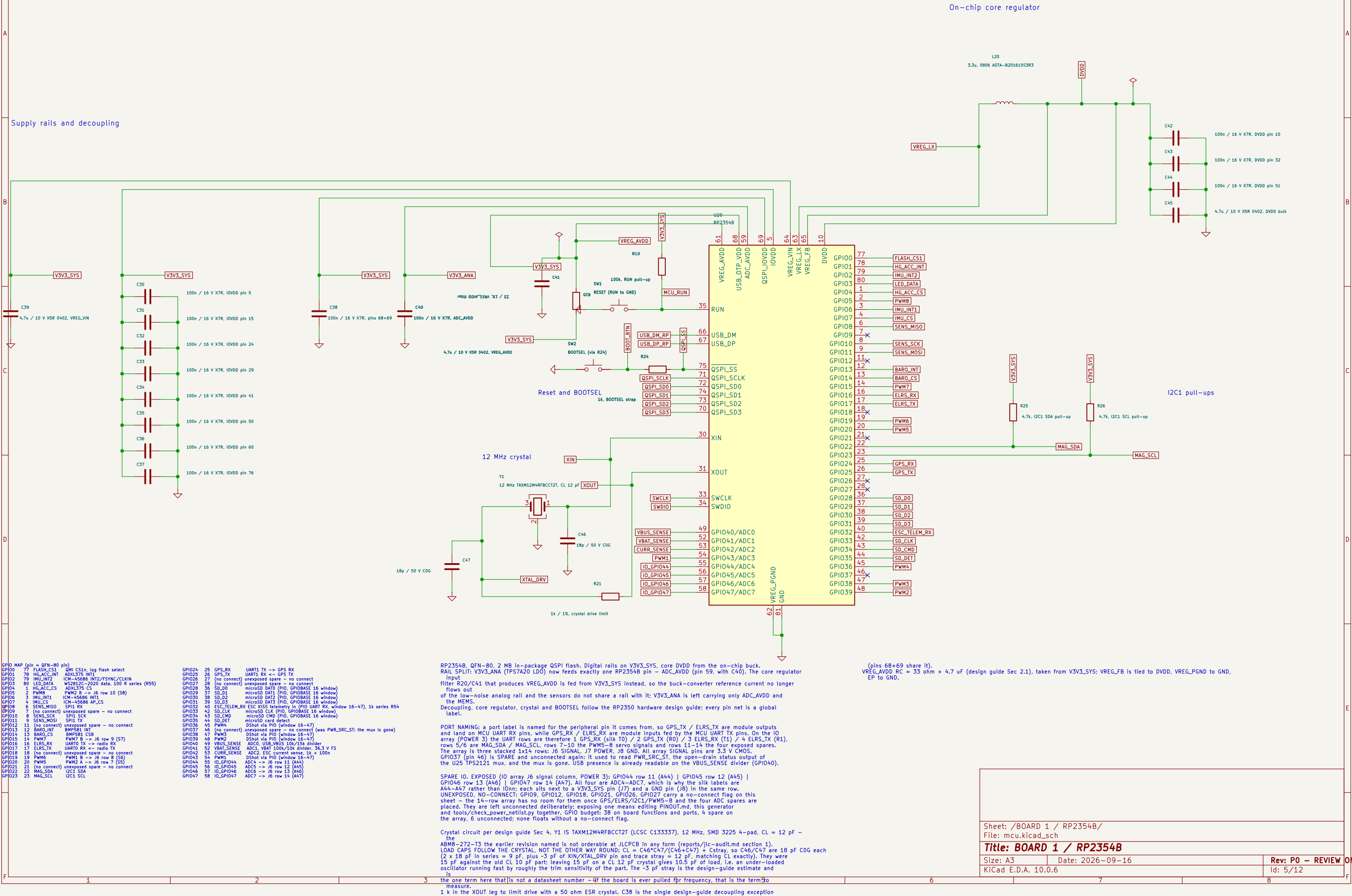


U7 AP63203WU–7, V5_SYS –> V3V3_SYS BUCK (Diodes D541326 Rev. 3–2). 3.8–32 V in, FIXED 3.3 V out, 2 A, 1.1 MHz with +/-6% frequency spread spectrum, 4 ms internal soft–start, 22 uA quiescent, TSOT26. It replaces the TP562913 two–stage stage (U7 + L2 + FB1 + a 0.1 % divider + five 22 uF) at ~8 parts instead of ~25. PASSIVES ARE TABLE 2 “Recommended Component Selections for AP63203” (3.3 V row) and Figure 21, the AP63203/AP63205 typical application, with one documented substitution:
L 3.9 uH in Table 2 –> 4.7 uH (L2). Sec 10 “Inductor” licenses it explicitly: “It is recommended to select an inductor of approximately 2.2uH to 10uH with a DC current rating of at least 35% higher than the maximum load current... Use a larger inductance for improved efficiency under light load conditions”, and this rail is a light load (~180 mA typical / ~300 mA peak). 4.7 uH is a part the board already sources.
C1 10 uF –> C8, 10 uF / 25 V X5R 0603, at VIN. C9 100 nF 0402 is an HF bypass beyond the datasheet, the same addition the removed AP63205 stage carried as C74.
C2 2 x 22 uF –> C10 / C11, 22 uF / 10 V X5R 0603 (GRM188R61A226ME15) at the output, exactly the table value.
C3 100 nF –> C13, bootstrap BST–SW.
FB (pin 1) is a SENSE input on the fixed–output parts and goes straight to V3V3_SYS – Sec 9 “Setting the Output Voltage”: only the adjustable AP63200/AP63201 take a divider (Eq.6). So there is no feedback divider, no 0.1 % resistor pair and no setpoint arithmetic on this board any more: VOUT is the part number.
EN (pin 2): “The EN pin is a high voltage pin and can be directly connected to VIN”, abs max 35 V, threshold 1.18 V rising / 1.10 V falling, internal 1.5 uA pull–up. R52 100k from V5_SYS is that connection with a series element, so a UVLO divider (Eq.1/Eq.2) or a start–delay cap (Eq.3) can be added later without touching copper. OPERATING POINT at VIN = 5.0 V, VOUT = 3.3 V, fsw = 1.1 MHz, Eq.7: dIL = 3.3 x 1.7 / (5.0 x 4.7u x 1.1M) = 0.22 A pk–pk (0.27 A at the ~20% inductance limit), so Eq.8 gives a 0.41 A inductor peak at the 0.3 A peak rail load. On USB, VIN falls to ~4.5 V behind D1 and dIL drops to 0.17 A; the part stays in CCM–adjacent light–load operation either way and neither case comes near the 2 A rating or the inductor.
DERATING CAVEAT: C8 and C10/C11 are the datasheet NOMINAL values and ceramic DC–bias derating is not in them – a 22 uF / 10 V 0603 at 3.3 V keeps roughly 60 %, so the 44 uF of COUT is ~26 uF effective. That is still above the 2 x 22 uF the table asks for in nominal terms and is the same derating the removed stage carried.
L2 CJIANG FTC40403054R7MGA (LCS C39676083), the inductor the removed AP63205 stage used as L3 – 4.7 uH +/-20%, DCR 41 mOhm typ / 46 max, Irms 4.3 A typ / 41 A worst case, Isat 7.0 A typ / 6.0 A worst case, 4.1 x 4.1 x 3.0 mm molded metal–composite, shielded (SZ CJIANG FTC series datasheet Rev 7.0 2025/11/05, Notes 3/4/7; the usable rating is the LESSER of Irms and Isat, so 4.0 A). At a 0.41 A peak that is a 10x margin – enormously over–rated for this rail, and deliberately so: it is a part already on the BOM and on the reel, and it costs 46 mOhm x 0.3 A^2 = 4 mW. IF THE 4 x 4 x 3.0 mm FOOTPRINT EVER HAS TO SHRINK, the FTC3030200 family (3 x 3 x 2.0, LCS C7423318 at 2.2 uH) is the same library footprint family and 2.2 uH is still inside the datasheet’s 2.2–10 uH window – at 0.47 A pk–pk ripple.
V3V3_ANA feeds ICM–45686, BMP581, ADXL375 and the RP2354B ADC_AVDD pin (100 nF at each pin on the MCU sheet): VREG_AVDD is filtered from V3V3_SYS instead, so the analog rail carries no core–regulator current. U12 EN (pin 3) is tied to V3V3_SYS itself.
the AP63203 has no power–good output, so the sequencing is done by the rail rather than by a status pin – the LDO enables when the buck output is up and drops with it, which is the same buck–before–LDO ordering PWR_GOOD used to give, at no parts.
What is lost is the “in regulation” qualifier: EN now releases at U12’s own ~0.9 V enable threshold on a rising V3V3_SYS.
not at the point a comparator declares the buck settled. For a 10–20 mA sensor rail behind a 4 ms soft–start that is a few hundred microseconds of difference, and the sensors are held in reset by firmware for far longer than that.
PASSIVE PACKAGES (DESIGN_SPEC “Physical design”): resistors are 0402, ceramics <= 4.7 uF are 0402 and the 10–22 uF ceramics are 0603. RESISTORS ARE 0402, NOT 0201: JLCPCB has no Basic 0201 resistor of any value, so every 0201 line carried a per–unique–part Extended setup fee – see reports/JLC–audit.md. 0402 makes most of the resistor values Basic and is hand–reworkable, which 0201 is not. THE 0.1 % PAIR IS GONE with the TP562913: the AP63203 is a fixed–output part and has no feedback divider, so no 0.1 % thin–film line is ordered at all. C10/C11 (AP63203 COUT) are 22 uF / 10 V X5R 0603 (GRM188R61A226ME15); at 3.3 V DC bias an X5R 22 uF 0603 keeps roughly 60 % of its nominal value, so the 44 uF nominal is ~26 uF effective. Do not read the effective value off the BOM. The 22 uF 0603 ceramics remain machine–place parts – JLC assembles them, a bench iron will not

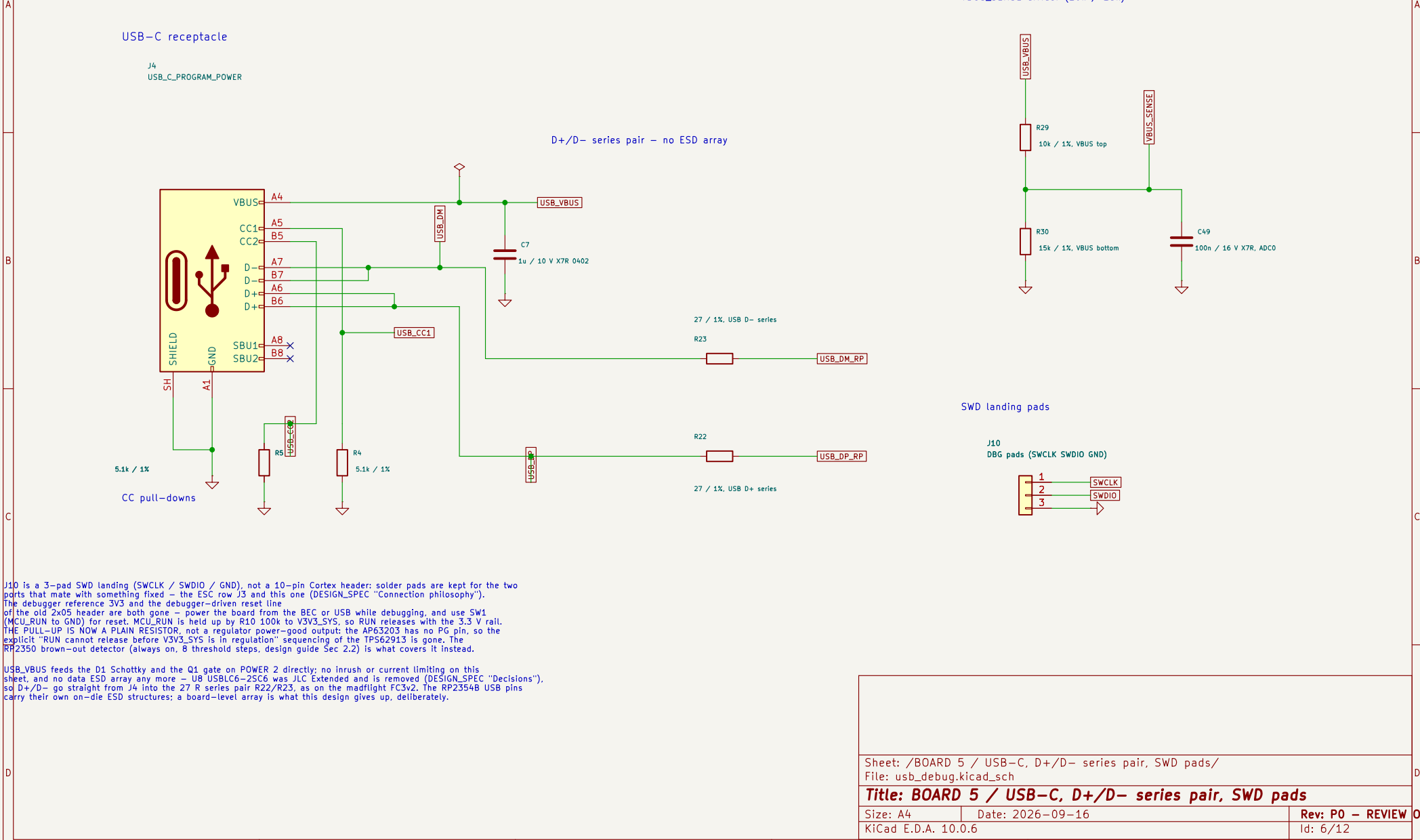
rework one.
C19, THE 5V_IN BULK, IS 100 uF / 6.3 V POLYMER IN AN EIA–3528–21 (B) CASE. It moved from V5_SYS to 5V_IN with the BEC decision; hold–up belongs on the rail the servo and module rows are on, which is now ahead of the OR–ing FET, and V5_SYS keeps only C8/C9 at the AP63203 input; ESR must stay <= 40 mOhm; Panasonic 6TPE100MAZB or a KEMET T520/T530 B case, 6.3 V on a 5.0 V rail is a 1.26x derating, the normal polymer figure; a tantalum electrolytic would need 2x and would not qualify.
THE ngspice DECKS IN simulations/ STILL MODEL THE OLD TOPOLOGY (a TPS2121 mux fed by an on–board pack buck, and a TP562913–class 3.3 V stage). They were not re–derived for the BEC + P–FET + AP63203 tree, so every number they produce is now history, not evidence – see DESIGN_SPEC “Open Items”.

Sheet: /POWER 3 / 3.3 V system buck and analog LDO/ File: power_3v3.kicad_sch		
Title: POWER 3 / 3.3 V system buck and analog LDO		
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
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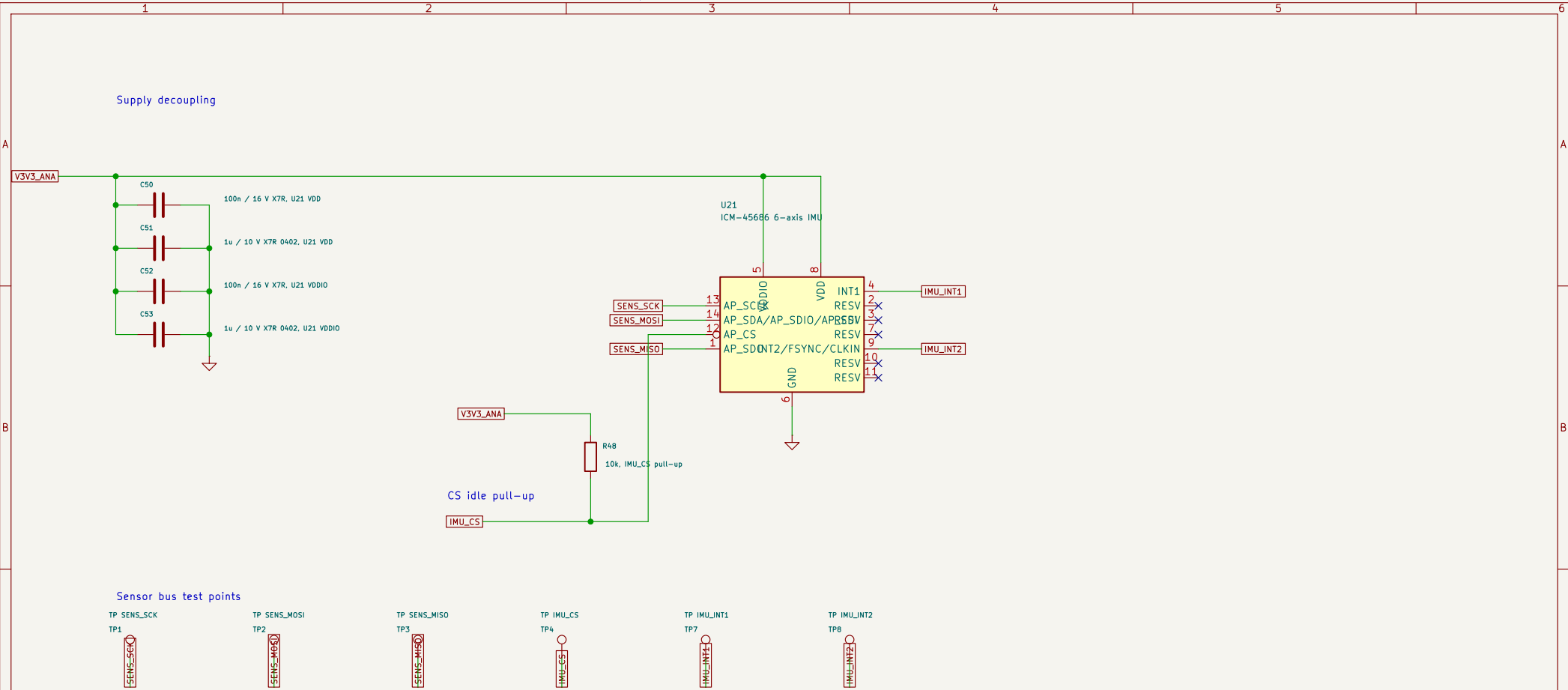
BOARD 1 – RP2354B, QFN–80, 2 MB IN–PACKAGE QSPI FLASH. Digital rails on V3V3_SYS, core DVDD from the on–chip buck. Supply pins fan out on the left into one decoupling group per rail.



AND THE J10 SW¹ LANDING PADS. NO ON-BOARD² USB ESD ARRAY – U8 USBLC6³ was removed.



BOARD 2 – U21 ICM–45686 6–axis IMU on the shared sensor SPI bus, powered from V3V3_ANA.



All three sensors run 4–wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM–45686 as a deliberate addition – its datasheet BOM (DS–000577 Rev 1.0 Table 11) lists 0.1 uF only.

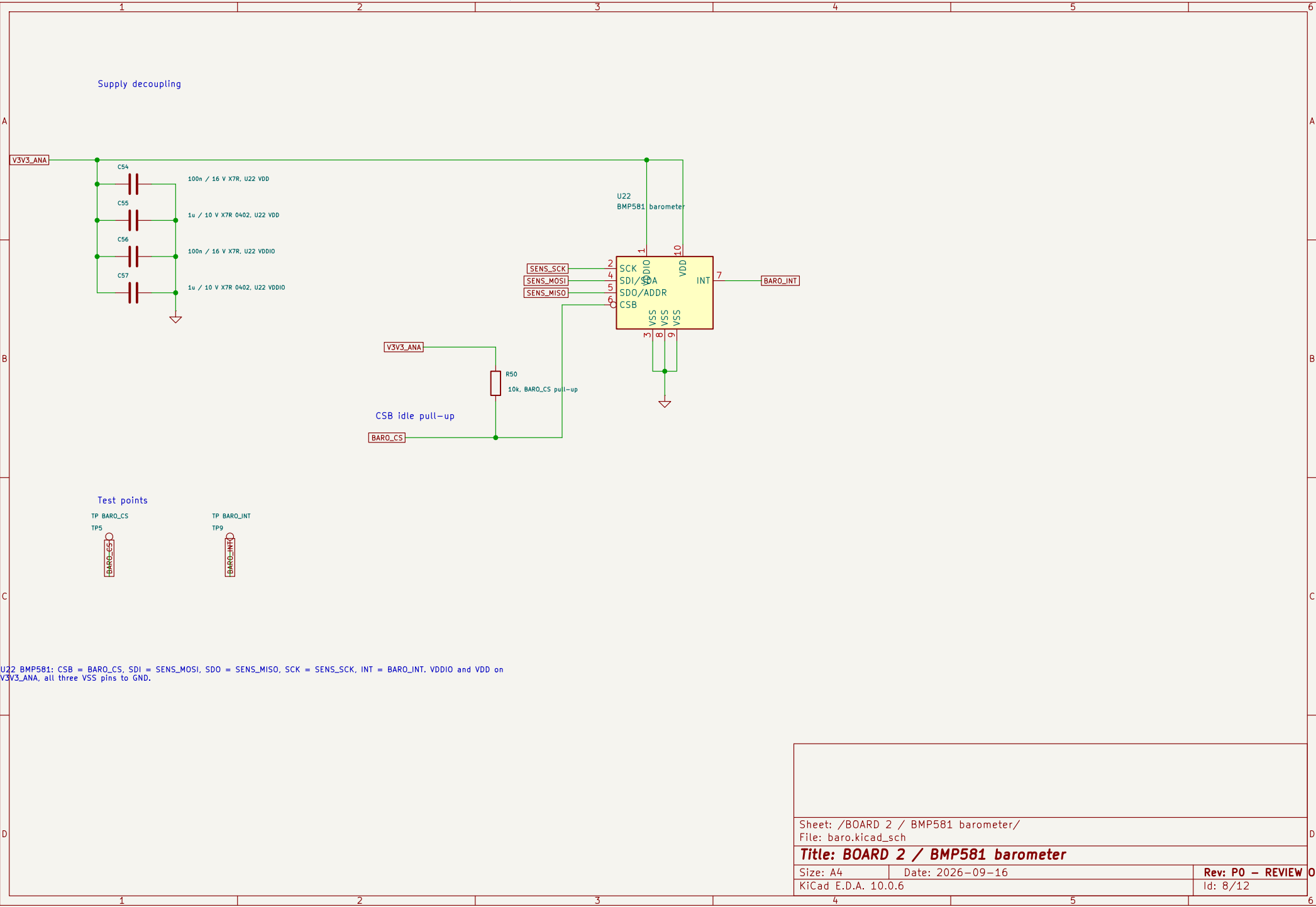
U21 ICM–45686 (TDK InvenSense DS–000577 Rev 1.0, Table 10 “Signal Descriptions” / Figure 4 pin–out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADD (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSYNC/CLKIN are host–driven inputs while INT2 is a chip output; only the INT2 function is used here, VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single–Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per–pin notes in Table 10); they are only needed for the AUX1 or I2C–master dual–interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 “Bill of Materials” (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight–controller practice.

R48, R50, R51: 10k pull–ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip–select pull–up) is removed along with that net – the ICM–45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull–up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

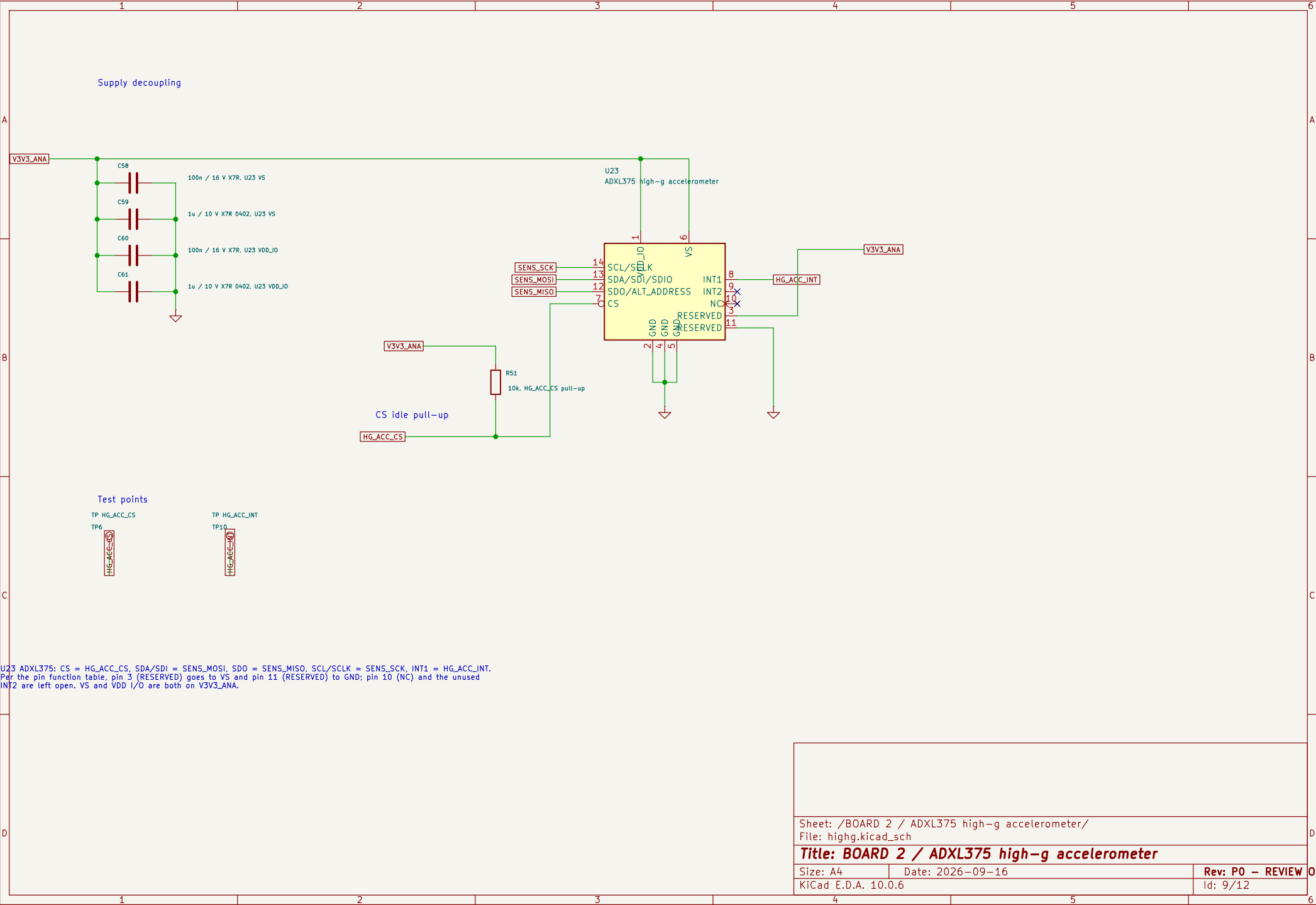
TEST POINTS TP1–TP10 (TestPoint:TestPoint_Pad_1.0x1.0mm, 1 x 1 mm bare F.Cu pad, no 3D model by design – tools/audit_footprints.py --no-3d-ok covers TestPoint_ along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else; at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.

Sheet: /BOARD 2 / ICM–45686 six–axis IMU/ File: imu.kicad_sch		
Title: BOARD 2 / ICM–45686 six–axis IMU		
Size: A4	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 7/12

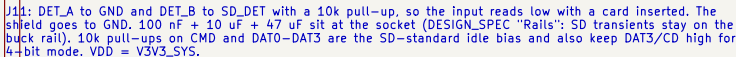
BOARD 2 – U22 BMP581 BAROMETER on the shared sensor SPI bus, powered from V3V3_ANA.



BOARD 2 – U23 ADXL375 HIGH-G ACCELEROMETER on the shared sensor SPI bus, from V3V3_ANA.



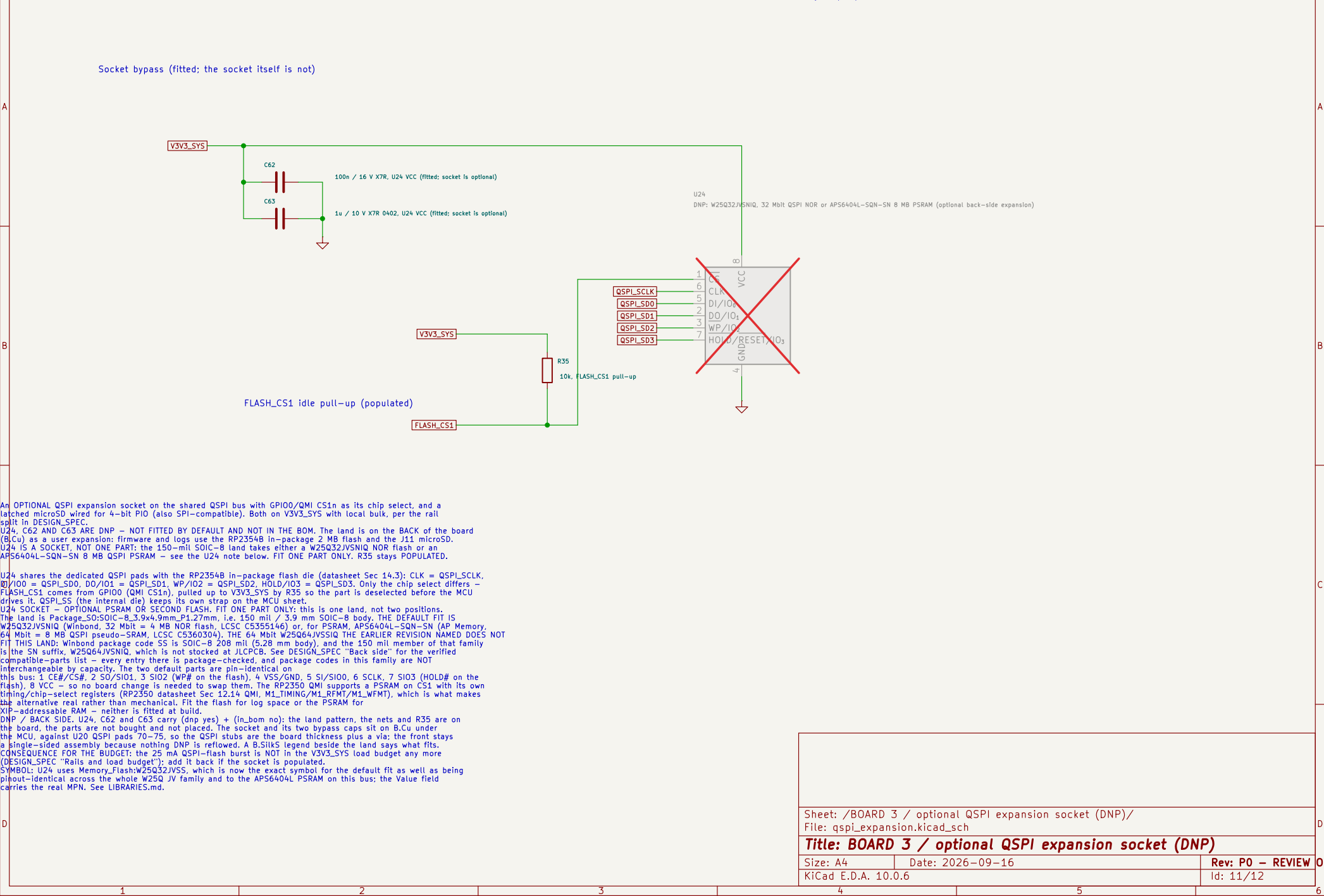
VDD = V3V3_SYS1 with 100 nF + 10 uF + 47 uF² at the socket; 10k idle pull-ups³ on CMD, DAT0-3 and card detect⁴.



Id: 10/12

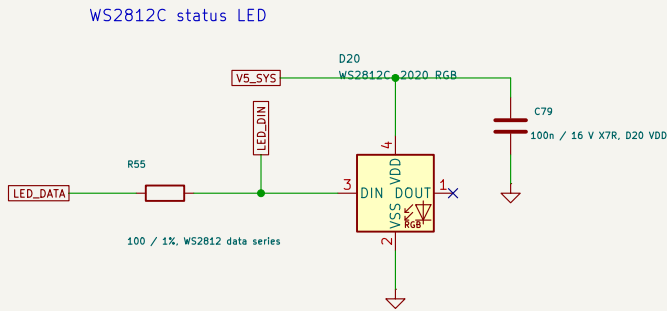
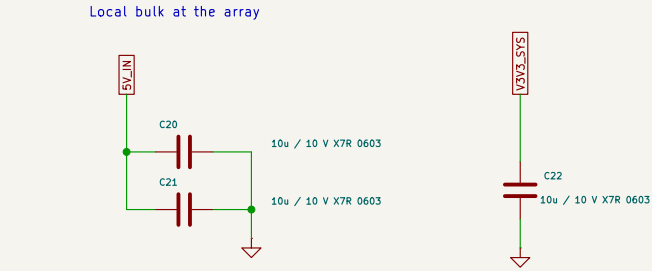
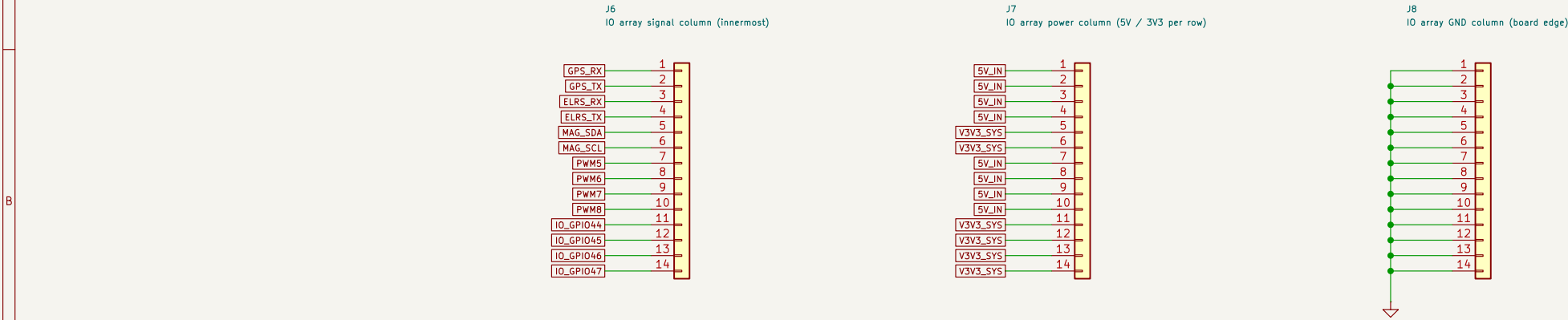
BOARD 3 – U24 OPTIONAL QSPI EXPANSION SOCKET, DNP, ON THE BACK (B.Cu).

One 150-mil SOIC-8 land: fit EITHER a W25Q32JVSNQ NOR flash OR an APS6404L-SQN-SN PSRAM. R35 stays populated.



BOARD 4 – THE 3 x 14 IO BLOCK (J6 SIGNAL / J7 POWER / J8 GND), THE WS2812C STATUS LED AND THE FOUR M3 GROMMET HOLES. Every signal that leaves the board except the ESC row J3 and the DBG pads J10 leaves here.

IO array – one row per module lead



IO ARRAY – ONE 3–COLUMN x 14–ROW THROUGH–HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflight FC3v2 style. It replaces the old 2x16 left–edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3–wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3–row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body–trimmed courtyard about exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS–009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) and rows 7–10 (servos S5–S8) take 5V_IN; rows 5–6 (magnetometer) and rows 11–14 (the four exposed spares) take V3V3_SYS. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis–plugged into a neighbour’s ground. THE EIGHT 5 V ROWS ARE ON 5V_IN, AHEAD OF THE OR–ING FET (DESIGN_SPEC “Decisions”, 2026–09–17). 5V_IN is the external BEC on J3 pin 7, so servo and module current goes from the BEC straight to the rows and never crosses Q1: the FET carries only the two 3.3 V regulators and the status LED, and a USB host never sees the servo rows at all. The rows are therefore NOT powered from USB – plugging in USB alone leaves every module and servo dead, which is the intended bench behaviour and the reverse of the previous revision. CURRENT: rows 1–4 and 7–10 count against the external BEC only; rows 5–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per–row current limit: the array is a system–integration connector, not a protected port.

MECHANICAL: H1–H4 are the four 4.0 mm grommet holes on the standard 30.5 x 30.5 mm pattern, off–centre in X, each now a PLATED hole with a 6.4 mm GND pad ring stitched by eight 0.6/0.3 mm vias, and a 6.5 mm top–side courtyard for the grommet flange. Pin 1 of every hole is GND, so the frame screws bond to the ground plane on every layer. See DESIGN_SPEC “Mounting and orientation”.

IO ARRAY ROW MAP (row n = J6 pin n signal | J7 pin n power | J8 pin n GND)

1	T0	GPS_RX	5V_IN	GND	UART1_TX → GPS_RX (GPIO24)
2	R0	GPS_TX	5V_IN	GND	UART1_RX ← GPS_TX (GPIO25)
3	T1	ELRS_RX	5V_IN	GND	UART0_TX → ELRS_RX (GPIO16)
4	R1	ELRS_TX	5V_IN	GND	UART0_RX ← ELRS_TX (GPIO17)
5	SDA	MAG_SDA	V3V3_SYS	GND	I2C1_SDA (GPIO22), 4.7k pull–up on the MCU sheet
6	SCL	MAG_SCL	V3V3_SYS	GND	I2C1_SCL (GPIO23), 4.7k pull–up on the MCU sheet
7	S5	PWM5	5V_IN	GND	PWM2 A (GPIO20), servo 5
8	S6	PWM6	5V_IN	GND	PWM1 B (GPIO15), servo 6
9	S7	PWM7	5V_IN	GND	PWM7 B (GPIO15), servo 7
10	S8	PWM8	5V_IN	GND	PWM2 B (GPIO5), servo 8
11	A44	IO_GPIO44	V3V3_SYS	GND	GPIO44 = ADC4, spare
12	A45	IO_GPIO45	V3V3_SYS	GND	GPIO45 = ADC5, spare
13	A46	IO_GPIO46	V3V3_SYS	GND	GPIO46 = ADC6, spare
14	A47	IO_GPIO47	V3V3_SYS	GND	GPIO47 = ADC7, spare

C20 / C21: 5V_IN local bulk at the array (rows 1–4 GPS/ELRS and rows 7–10 servo, the 5 V supply rows). C22: V3V3_SYS local bulk at the array (rows 5–6 and 11–14).

PWM MAP
PWM5 = GPIO20 (row 7, S5)
PWM6 = GPIO19 (row 8, S6)
PWM7 = GPIO15 (row 9, S7)
PWM8 = GPIO5 (row 10, S8)

ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO43

Sheet: /BOARD 4 / IO array, status LED, mounting holes/
File: io_block.kicad_sch

Title: BOARD 4 / IO array, status LED, mounting holes

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026–09–16

Rev: P0 – REVIEW ONLY

Id: 12/12